

T3 matched BSM $\Delta\mathcal{L}_{\text{EFT}}$ terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from matched BSM EFT terms. Empty cells denote no matched term with that field combination.

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	$-\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$ $-\frac{5}{12} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) B_{\mu_1 \mu_2}^2$	$+C_{H2} H_{a_1} H_{a_1}^\dagger$ $-H_{a_1} \mu^2 H_{a_1}^\dagger$ $+\frac{3\text{lambda}dH\text{Inv}2^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{15\text{lambda}dH\text{Inv}1^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{14\epsilon}$ $+\frac{95\text{lambda}dH\text{Inv}2^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{42\epsilon}$ $+\frac{10\sqrt{\frac{2}{3}}\text{lambda}dH\text{Inv}1\text{lambda}dH\text{Inv}2h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{7\epsilon}$ $+\frac{3}{2}\text{lambda}dH\text{Inv}2^2 h \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{15}{14}\text{lambda}dH\text{Inv}1^2 h \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{95}{42}\text{lambda}dH\text{Inv}2^2 h \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{10}{7}\sqrt{\frac{2}{3}}\text{lambda}dH\text{Inv}1\text{lambda}dH\text{Inv}2h \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2}$ $+\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) \lambda_{T3}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2}$ $-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2}$ $-\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) \lambda_{T3}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2}$	$-\frac{6\hbar \left(W_{\mu_1 \mu_2}^{A_1} \right)^2}{\epsilon}$ $-5\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$ $-\frac{1}{6}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$ $-\frac{5}{6}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$-\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T(1,1)W_{T(2,1)T(4,1)2}^T} C P_6 \ell_{a_2,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,2} \ell_{a_2,4} M_F^2}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}$ $+\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T(1,1)W_{T(2,1)T(4,1)2}^T} C P_6 \ell_{a_2,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,2} \ell_{a_2,4} M_F^2}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}$ $-\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T(1,1)W_{T(2,1)T(4,1)2}^T} C P_6 \ell_{a_2,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,2} \ell_{a_2,4} M_F^2}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}$ $+\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T(1,1)W_{T(2,1)T(4,1)2}^T} C P_6 \ell_{a_2,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,2} \ell_{a_2,4} M_F^2}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}$ $-\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T(1,1)W_{T(2,1)T(4,1)2}^T} C P_6 \ell_{a_2,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,2} \ell_{a_2,4} M_F^2}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}$ $+\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T(1,1)W_{T(2,1)T(4,1)2}^T} C P_6 \ell_{a_2,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,2} \ell_{a_2,4} M_F^2}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}$ $-\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T(1,1)W_{T(2,1)T(4,1)2}^T} C P_6 \ell_{a_2,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,2} \ell_{a_2,4} M_F^2}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}$ $+\frac{2\sqrt{\frac{2}{3}}h\log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T(1,1)W_{T(2,1)T(4,1)2}^T} C P_6 \ell_{a_2,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,2} \ell_{a_2,4} M_F^2}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	e
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